

► Cloud computing has several economic motivations:

1. Utility Pricing

- Cloud service don't need to be cheaper to be economic
- Whether a solution is economic depends on how to service the demand

2. Reducing capital expenditures, increasing operational expenses

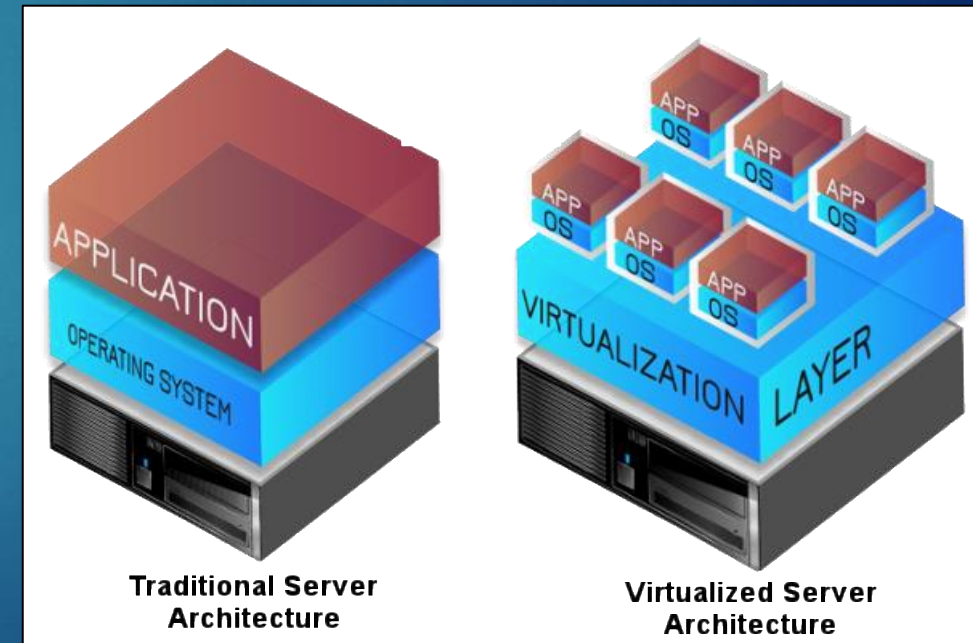
- CAPEX requires a large up-front investment to be amortized (written down) as its value decreases over time
- OPEX is a day-to-day expense which can much more easily be increased or decreased as business needs dictate

3. Efficiently allocating resources

- Cloud computing provides the ability to more efficiently allocate compute and storage resources vs dedicated server approaches
- This translates into more direct control over how business use cloud services that can result in more efficient business operations and ultimately more costs savings

► Virtualization

- Refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources
- The virtualization layer is referred to as a hypervisor, which is a program that allows multiple operating systems to share a single hardware host
- Two major types: Hosted and Native
- Advantages:
 - More efficient use of hardware, provisioning fast and simpler, easier to maintain
- Disadvantages:
 - Virtualization software, licensing, security must protect both physical and virtual machines



AWS Overview

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology



AWS Console

4

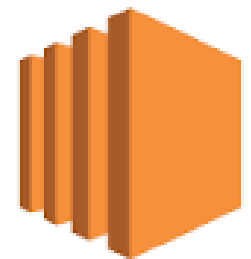
The **AWS Management Console** is a web application that comprises and refers to a broad collection of service consoles for managing AWS resources. When you first sign in, you see the console home page. The home page provides access to each service console and offers a single place to access the information you need to perform your AWS related tasks

- ▶ Over 200 services available
- ▶ Customizable interface (e.g. shortcuts)
- ▶ Build and monitor applications
- ▶ Monitor costs and set alerts
- ▶ Aside from a web browser, you can also access it through the mobile app version
- ▶ Let's take a tour...



Last class you created a virtual server...

- ▶ EC2 (Amazon's Elastic Cloud Computing)
 - ▶ Provides the ability for users to create virtual machines as needed (elastic)
 - ▶ Users are billed calculated by either the hour or the second based on the size of the instance, operating system, and the Region
 - ▶ If you STOP an instance, you will not be charged while it is in a STOPPED state BUT you will for any attached storage since it is persisted
 - ▶ The larger the server, the more cost per/hour
- ▶ There are several options you can choose when you create an EC2 instance
 - ▶ Let's walkthrough them...

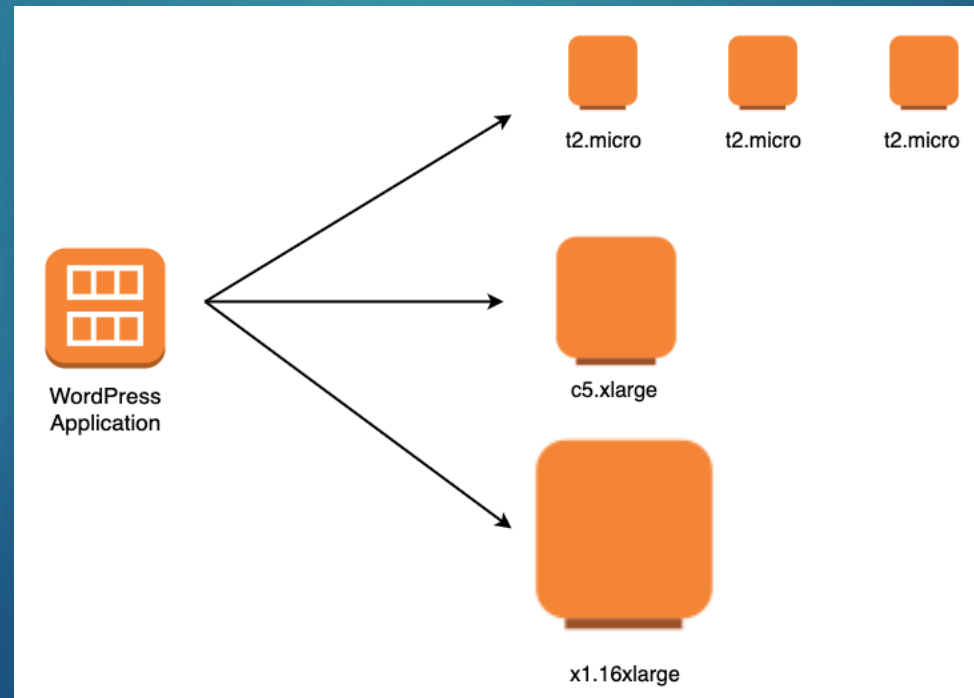


Amazon
EC2

EC2

Amazon Machine Image (AMI)

- ▶ AMI is the “Golden Master”
- ▶ It provides the information required to launch an instance of any size
- ▶ You can launch multiple instances from a single AMI when you need multiple instances with the same configuration



Amazon Machine Image (AMI)

- ▶ Virtual machines are created from templates known as an Amazon Machine Image (AMI)
- ▶ It provides the information required to launch an instance

The screenshot displays the AWS IAM console's AMI selection interface. At the top, four categories are listed: 'Common AMIs for Linux and Windows', 'Your custom AMIs that can be shared/reused', '3rd-party AMIs with pre-installed software and verified by AWS', and 'Comes from AWS users, and are not verified by AWS'. Below these, the console shows 'Quickstart AMIs (45)', 'My AMIs (1)', 'AWS Marketplace AMIs (5764)', and 'Community AMIs (500)'. The 'Refine results' sidebar on the left includes a 'Clear all filters' button and a 'Free tier only' filter that is checked. Under 'OS category', 'All Linux/Unix' is selected. Under 'Architecture', '64-bit (x86)' is selected. The main area shows a list of AMIs, including 'Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type' and 'Red Hat Enterprise Linux 8 (HVM), SSD Volume Type'. Each entry includes the AMI ID, a description, platform, root device type, virtualization type, and ENA status. A callout box on the left points to the 'Free tier only' filter with the text 'Check this to view only free tier options'.

Common AMIs for Linux and Windows

Your custom AMIs that can be shared/reused

3rd-party AMIs with pre-installed software and verified by AWS

Comes from AWS users, and are not verified by AWS

Quickstart AMIs (45)
Commonly used AMIs

My AMIs (1)
Created by me

AWS Marketplace AMIs (5764)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

☒ Free tier only Info

OS category

☐ All Linux/Unix

☐ All Windows

Architecture

☐ 64-bit (Arm)

☐ 32-bit (x86)

☐ 64-bit (x86)

☐ 64-bit (Mac)

☐ 64-bit (Mac-Arm)

All products (16 filtered, 45 unfiltered)

aws Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type
ami-0cff7528ff583bf9a (64-bit (x86)) / ami-00bf5f1c358708486 (64-bit (Arm))
Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.
Platform: amazon Root device type: ebs Virtualization: hvm ENA enabled: Yes
☒ 64-bit (x86) ☐ 64-bit (Arm)

aws Amazon Linux 2 AMI (HVM) - Kernel 4.14, SSD Volume Type
ami-065efef2c739d613b (64-bit (x86)) / ami-09f0bb50202ca06b0 (64-bit (Arm))
Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.
Platform: amazon Root device type: ebs Virtualization: hvm ENA enabled: Yes
☒ 64-bit (x86) ☐ 64-bit (Arm)

Red Hat Red Hat Enterprise Linux 8 (HVM), SSD Volume Type
ami-06640050dc3f556bb (64-bit (x86)) / ami-0238411fb452f8275 (64-bit (Arm))
Red Hat Enterprise Linux version 8 (HVM), EBS General Purpose (SSD) Volume Type
Platform: rhel Root device type: ebs Virtualization: hvm ENA enabled: Yes
☒ 64-bit (x86) ☐ 64-bit (Arm)

SUSE SUSE Linux Enterprise Server 15 SP3 (HVM), SSD Volume Type
ami-08895422b5f3aa64a (64-bit (x86)) / ami-08f182b25f271ef79 (64-bit (Arm))
SUSE Linux Enterprise Server 15 Service Pack 3 (HVM), EBS General Purpose (SSD) Volume Type. Amazon EC2 AMI Tools preinstalled; Apache 2.2, MySQL 5.5, PHP 5.3, and Ruby 1.8.7 available.
Platform: suse Root device type: ebs Virtualization: hvm ENA enabled: Yes
☒ 64-bit (x86) ☐ 64-bit (Arm)

Check this to view only free tier options

EC2 Instances - Sharing

- ▶ So, if you are building an application on an EC2 instance, can you share this with your team members?
 - ▶ You cannot share a specific EC2 instance with your team



- ▶ You can share an AMI image

Creating an AMI Image

- ▶ Using your completed WordPress EC2 instance, you can create an AMI image that can be reused
- ▶ To do this, select **Actions > Images and Templates > Create Image**

The screenshot shows the AWS Management Console 'Instances' page. A single instance named 'WordPress' is listed with ID 'i-0a13cc466f3ecd...', state 'Running', and type 't2.micro'. The 'Status check' shows 'Initializing'. The 'Actions' dropdown menu is open, displaying options like 'Connect', 'View details', and 'Image and templates'. The 'Image and templates' option is selected, opening a sub-menu where 'Create image' is highlighted.

	Name	Instance ID	Instance state	Instance type	Status check
☒	WordPress	i-0a13cc466f3ecd...	Running	t2.micro	Initializing

Creating an AMI Image

10

#1 - Name your image and select "Create image"

Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Instance ID
i-0a13cc466f3ecdd89

Image name

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Maximum 255 characters

No reboot
☐ Enable

No tags associated with the resource.

You can add 50 more tags.

#2 - The image is now available under:
Images > AMIs

New EC2 Experience

Launch EC2 Image Builder Actions

Owned by me Filter by tags and attributes or search by keyword

Name	AMI Name	AMI ID	Source	Owner	Visibility	Status
	WordPress	ami-026478a127382b4e3	075837463923/...	075837463923	Private	available

Image: ami-026478a127382b4e3

Details Permissions Tags

AMI ID	ami-026478a127382b4e3	AMI Name	WordPress
Owner	075837463923	Source	075837463923/WordPress
Status	available	State Reason	-

Edit

Creating an AMI Image

- ▶ Your image is now available under “My AMIs” when you configure an EC2 instance to launch

Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Quickstart AMIs (45)
Commonly used AMIs

My AMIs (2)
Created by me

AWS Marketplace AMIs (5913)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

▼ Owner

☒ Owned by me
☐ Shared with me

▼ OS category

☐ All Linux/Unix
☐ All Windows

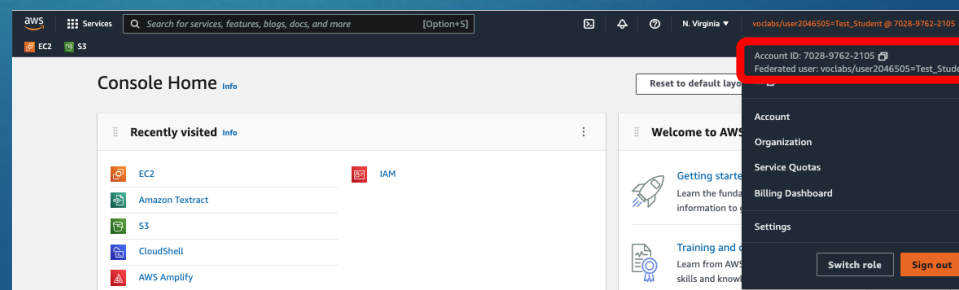
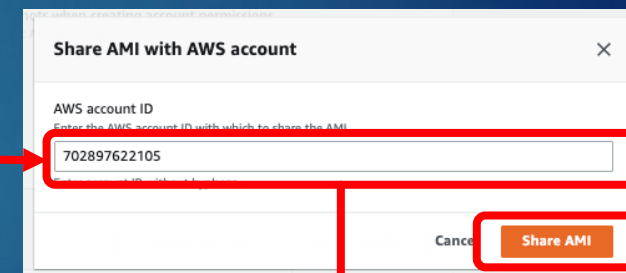
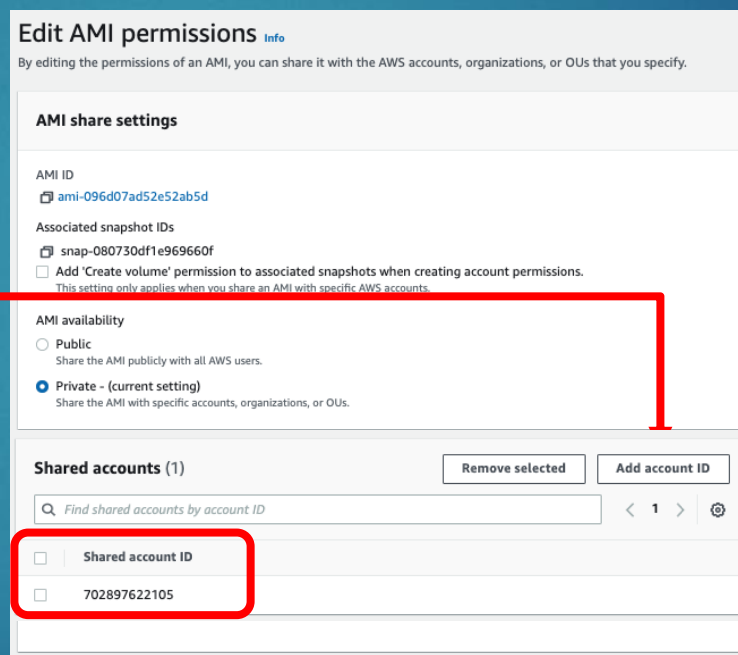
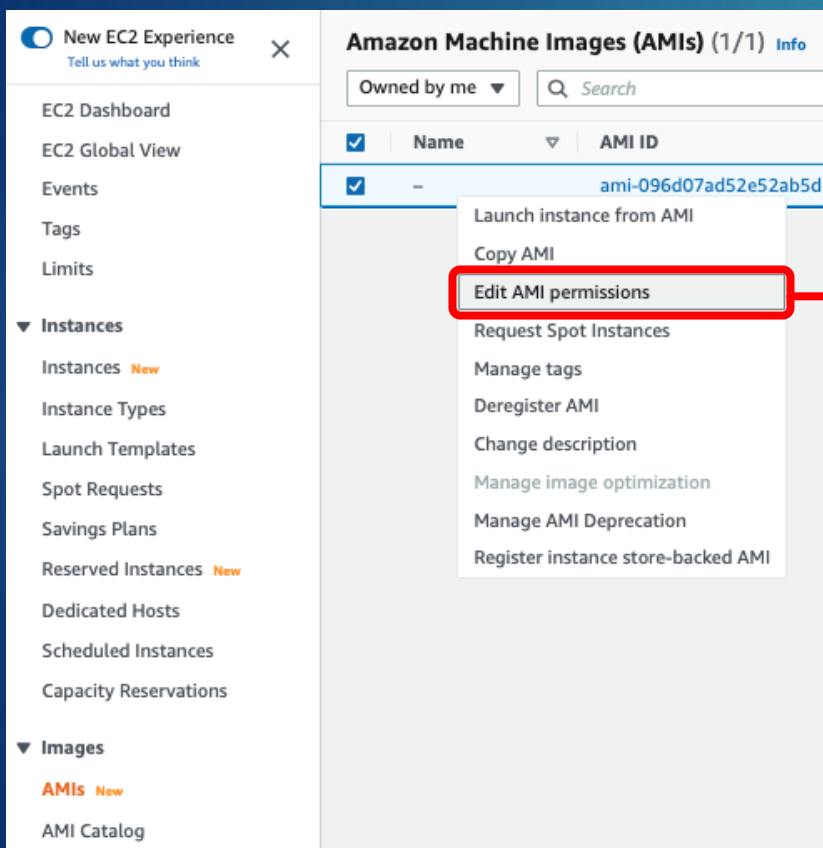
All products (1 filtered, 2 unfiltered)

 **WordPress**
ami-096d07ad52e52ab5d
Platform: Other Linux Architecture: x86_64 Owner: 702897622105 Publish date: 2022-08-06 Root device type: ebs Virtualization: hvm ENA enabled: Yes

Select

Sharing an AMI Image

- ▶ Under AMI, Select AMI image and select **Edit AMI permissions**
- ▶ Click “Add account ID” and add the account # of your team members and click “Share AMI”



Create Instance with a Shared AMI

- ▶ Team members that had their account # added, the shared image will appear under “My AMIs” when you create an EC2 instance

Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Search for an AMI by entering a search term e.g. "Windows"

Quickstart AMIs (45)
Commonly used AMIs

My AMIs (2)
Created by me

AWS Marketplace AMIs (5913)
AWS & trusted third-party AMIs

Community AMIs (500)
Published by anyone

Refine results

Clear all filters

▼ Owner

☐ Owned by me

☒ Shared with me

OS category

Linux/Unix

All products (1 filtered, 2 unfiltered)

Test image - MZ

ami-077d3e095947767e5

Platform: Other Linux Architecture: x86_64 Owner: 479985889380 Publish date: 2022-07-24 Root device type: ebs ENA enabled: Yes

Select

Make sure you check the “Shared with me” checkbox

AMI and OS

- ▶ Available in Windows, Linux and Mac
 - ▶ In class, will be working exclusively in Linux
 - ▶ For your team project, you can pick whatever OS you want, but be aware they are all not free
- ▶ There are several Linux OS available
 - ▶ These include Redhat, Ubuntu and SUSE
- ▶ We have been using Amazon Linux AMI
 - ▶ Integrates easier with AWS services and tools
 - ▶ Optimized for use in Amazon EC2 with a latest and tuned Linux kernel version



Amazon Machine Image – Many Instance configurations to choose from

► From small virtual servers...

	Instance type ▾	vCPUs ▾	Architecture ▾	Memory (GiB) ▾	Storage (GB) ▾	Storage type ▾	Network performance ▾	On-Demand Linux pricing ▾	On-Demand Windows pricing
☐	t1.micro	1	i386, x86_64	0.612	-	-	Very Low	0.02 USD per Hour	0.02 USD per Hour
☐	t2.nano	1	i386, x86_64	0.5	-	-	Low to Moderate	0.0058 USD per Hour	0.0081 USD per Hour
☒	t2.micro	1	i386, x86_64	1	-	-	Low to Moderate	0.0116 USD per Hour	0.0162 USD per Hour
☐	t2.small	1	i386, x86_64	2	-	-	Low to Moderate	0.023 USD per Hour	0.032 USD per Hour
☐	t2.medium	2	i386, x86_64	4	-	-	Low to Moderate	0.0464 USD per Hour	0.0644 USD per Hour

To very large...and pricey!

☐	u-6tb1.56xlarge	224	x86_64	6144	-	-	100 Gigabit	46.40391 USD per Hour	56.70791 USD per Hour
☐	u-6tb1.112xlar...	448	x86_64	6144	-	-	100 Gigabit	54.6 USD per Hour	75.208 USD per Hour
☐	u-9tb1.112xlar...	448	x86_64	9216	-	-	100 Gigabit	81.9 USD per Hour	102.508 USD per Hour
☐	u-12tb1.112xl...	448	x86_64	12288	-	-	100 Gigabit	109.2 USD per Hour	129.808 USD per Hour

EC2 Instance Types

- Deciding which to use depends on what you need to do...

	Instance type ▾	vCPUs ▾	Architecture ▾
☐	t1.micro	1	i386, x86_64
☐	t2.nano	1	i386, x86_64
☒	t2.micro	1	i386, x86_64
☐	t2.small	1	i386, x86_64
☐	t2.medium	2	i386, x86_64

Family	Speciality	Use case
F1	Field Programmable Gate Array	Genomics research, financial analytics, real-time video processing, big data etc
I3	High Speed Storage	NoSQL DBs, Data Warehousing etc
G3	Graphics Intensive	Video Encoding/ 3D Application Streaming
H1	High Disk Throughput	MapReduce-based workloads, distributed file systems such as HDFS and MapR-FS
T2	Lowest Cost, General Purpose	Web Servers/Small DBs
D2	Dense Storage	Fileservers/Data Warehousing/Hadoop
R4	Memory Optimized	Memory Intensive Apps/DBs
M5	General Purpose	Application Servers
C5	Compute Optimized	CPU Intensive Apps/DBs
P3	Graphics/General Purpose GPU	Machine Learning, Bit Coin Mining etc
X1	Memory Optimized	SAP HANA/Apache Spark etc

Network settings – More next class!

The screenshot shows the 'Network settings' page for an AWS instance. It includes sections for Network, Subnet, Auto-assign public IP, and Firewall (security groups). Callouts explain that 'VPC = Virtual Private Cloud' points to the network ID, 'Subnet = IP Addresses' points to the subnet selection, and the firewall rules 'Controls the traffic to and from your instance'. Another callout points to the 'Allow HTTPs traffic from the internet' checkbox, stating 'Check to allow HTTP/HTTPS access (e.g. web site)'. A warning at the bottom states: 'Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

Network settings [Get guidance](#) [Edit](#)

Network [Info](#)
vpc-055d45f9bdb0fb33c ← VPC = Virtual Private Cloud

Subnet [Info](#)
No preference (Default subnet in any availability zone) ← Subnet = IP Addresses

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance. ← Controls the traffic to and from your instance

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-4' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance Anywhere
0.0.0.0/0

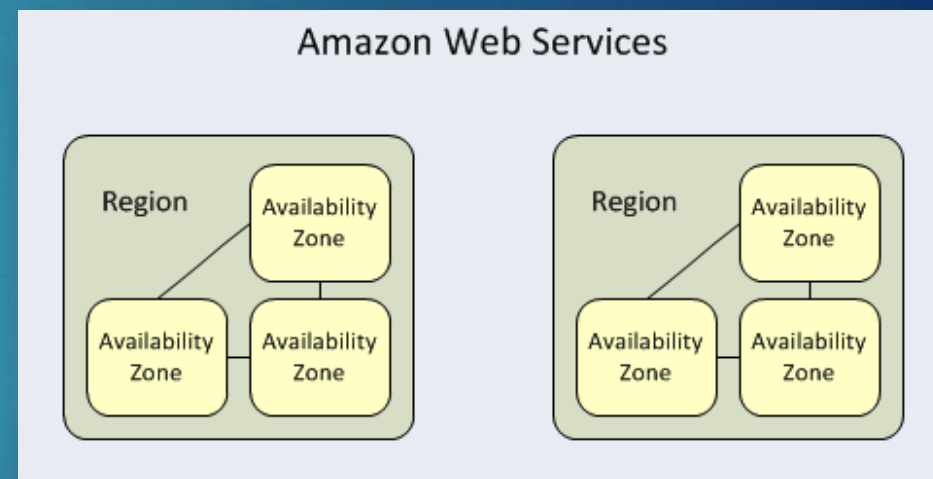
☐ Allow HTTPs traffic from the internet
To set up an endpoint, for example when creating a web server ← Check to allow HTTP/HTTPS access (e.g. web site)

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

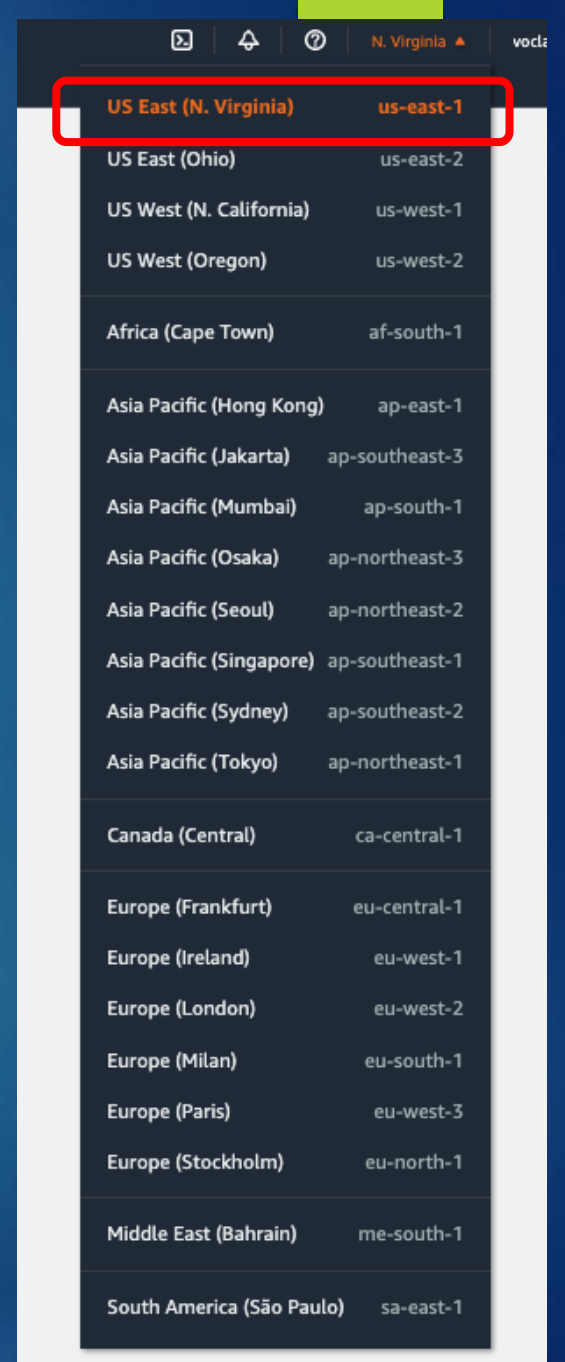
Regions

- ▶ Each region represents a separate geographic area, where your EC2 instances are running
- ▶ It has multiple, isolated locations known as Availability Zones
- ▶ AWS resources (e.g. instances, key pairs, storage) are either:
 - ▶ Global
 - ▶ Tied to a Region
 - ▶ Tied to an Availability Zone
- ▶ Each Region is designed to be completely isolated from the other to provide fault tolerance and stability



Regions

- ▶ Why is this important?
 - ▶ Certain resources might not be available if created in different regions or availability zones
 - ▶ For example, your key pair is tied to the region it was created in
 - ▶ You cannot create an EC2 instance in another region with the same key pair unless you upload to that specific region
 - ▶ Another important factor is costs across regions
 - ▶ Any idea which is the most expensive?



The screenshot shows the AWS Management Console interface with the 'Regions' page selected. The 'US East (N. Virginia)' region is highlighted with a red box. The table lists various AWS regions and their corresponding availability zones.

Region	Availability Zone
US East (N. Virginia)	us-east-1
US East (Ohio)	us-east-2
US West (N. California)	us-west-1
US West (Oregon)	us-west-2
Africa (Cape Town)	af-south-1
Asia Pacific (Hong Kong)	ap-east-1
Asia Pacific (Jakarta)	ap-southeast-3
Asia Pacific (Mumbai)	ap-south-1
Asia Pacific (Osaka)	ap-northeast-3
Asia Pacific (Seoul)	ap-northeast-2
Asia Pacific (Singapore)	ap-southeast-1
Asia Pacific (Sydney)	ap-southeast-2
Asia Pacific (Tokyo)	ap-northeast-1
Canada (Central)	ca-central-1
Europe (Frankfurt)	eu-central-1
Europe (Ireland)	eu-west-1
Europe (London)	eu-west-2
Europe (Milan)	eu-south-1
Europe (Paris)	eu-west-3
Europe (Stockholm)	eu-north-1
Middle East (Bahrain)	me-south-1
South America (São Paulo)	sa-east-1

Regions – US East vs South America

- ▶ Running the same EC2 instance in different regions can vary in costs

Region: Operating system:

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t4g.nano	\$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	\$0.0047	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.nano	\$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t2.nano	\$0.0058	1	0.5 GiB	EBS Only	Low
t4g.micro	\$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	\$0.0094	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.micro	\$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit

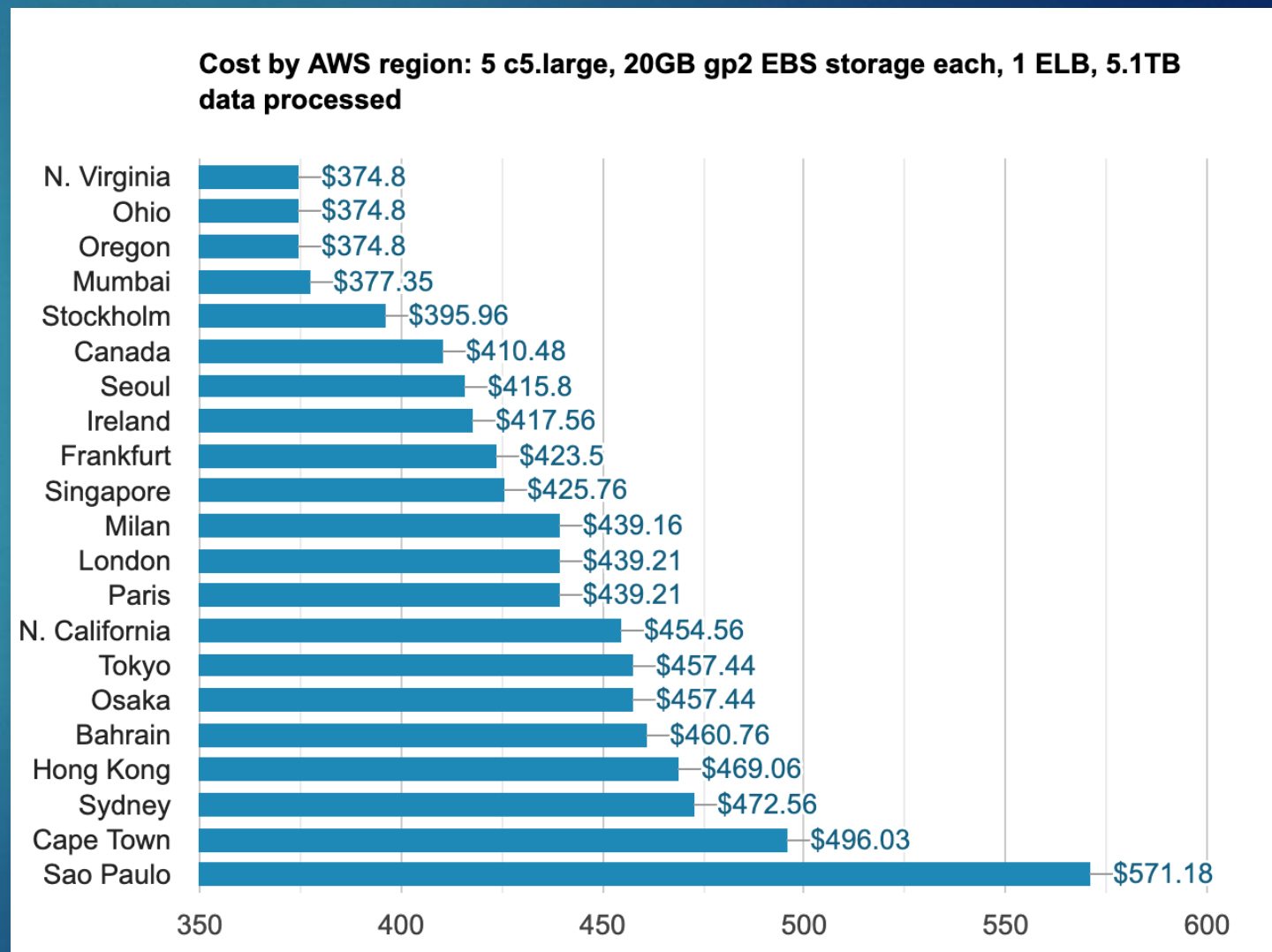
Region: Operating system:

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t4g.nano	\$0.0067	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	\$0.0076	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.nano	\$0.0084	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t2.nano	\$0.0093	1	0.5 GiB	EBS Only	Low
t4g.micro	\$0.0134	2	1 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	\$0.0151	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.micro	\$0.0168	2	1 GiB	EBS Only	Up to 5 Gigabit

- ▶ The same rules for costs apply to storage and data transfer

Regions – Side by side comparison

- ▶ Running the same hardware in Sao Paulo (Brazil) is **59% higher** than N. Virginia



Storage

- ▶ Each instance has the option to add different types and sizes of storage
- ▶ EBS (Elastic Block Storage) volumes persist independently from the running life of EC2 instances
 - ▶ Why is this important?
- ▶ By default, EBS disk is deleted when the associated instance is terminated

The screenshot displays the AWS Management Console interface for configuring an EBS volume. The page title is "Storage (volumes)" with an "Info" link and a "Simple" view toggle. Below this is the "EBS Volumes" section with a "Hide details" link. The configuration is for "Volume 1 (AMI Root)".

Storage type Info	Device name - required Info	Snapshot Info
EBS	/dev/xvda	snap-096dd96dde7d24189
Size (GiB) Info	Volume type Info	IOPS Info
8	gp3	3000
Delete on termination Info	Encrypted Info	KMS key Info
Yes	Not encrypted	Select
Throughput Info		
125		

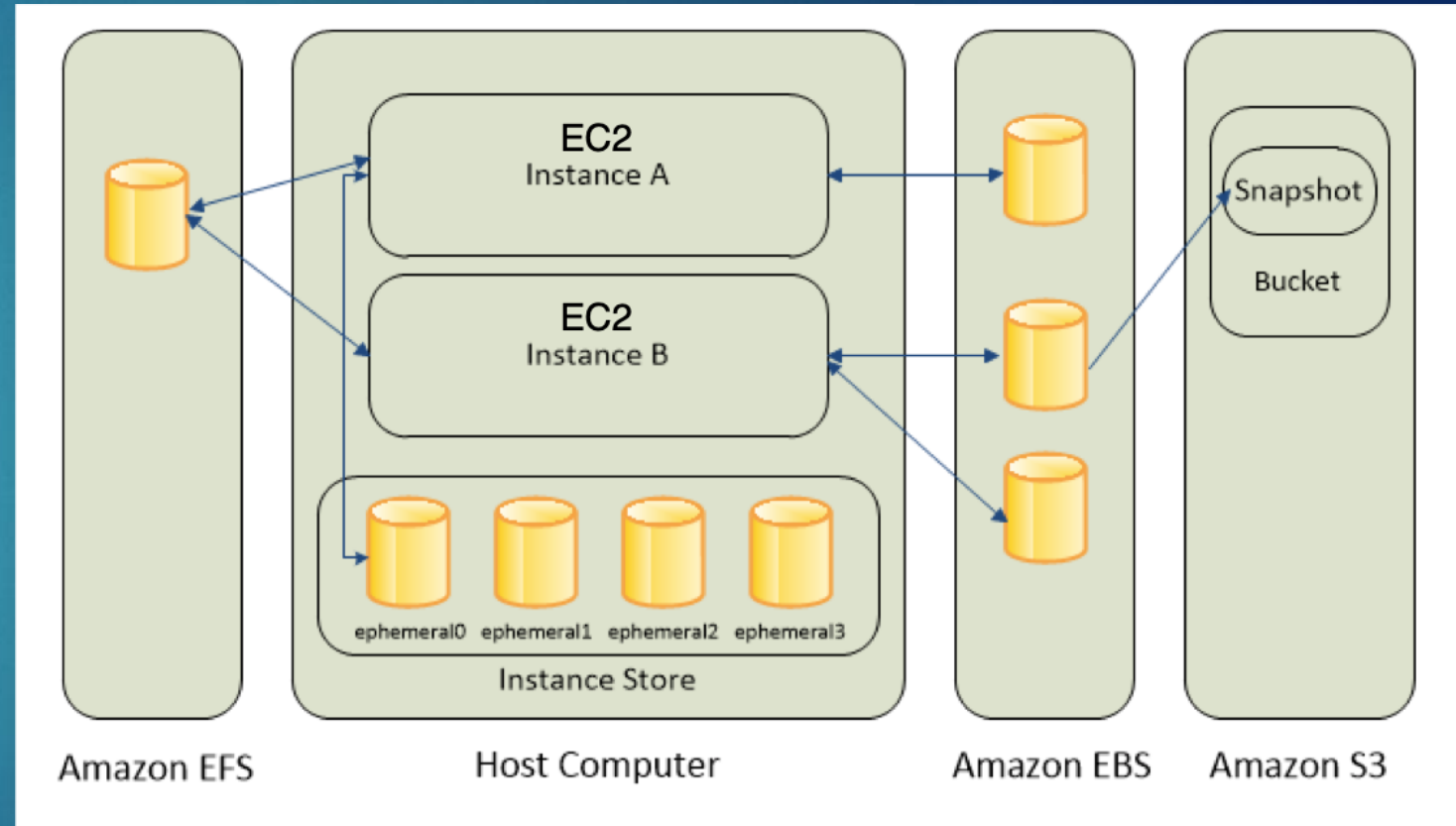
At the bottom, there is an "Add new volume" button. A note at the bottom right states: "KMS keys are only applicable when encryption is set on this volume."

Storage - Details

► Amazon EC2 offers 4 main storage options:

1. Amazon EC2 Instance Store
2. Amazon Simple Storage Service (S3)
3. Amazon Elastic Block Store (EBS)
4. Amazon Elastic File System (EFS)

Mounted to EC2 instance and exist independently



EC2 Instance Store

- ▶ Local to the instance only
- ▶ Provides very high random I/O performance, runs on SSD or HDD
- ▶ Non-persistent data store and data is lost under any of the following circumstances:
 - ▶ The underlying disk drive fails
 - ▶ The instance stops
 - ▶ The instance terminates
- ▶ Ideal for temporary storage of information that changes frequently, such as buffers, caches, scratch data, and other temporary content



Storage – Simple Storage Service (S3)

- ▶ Provides simple object storage, useful for hosting simple websites with images and videos
- ▶ Every object stored in Amazon S3 is contained in a “bucket”
 - ▶ S3 buckets are created in a Region within a partition (group of regions)
 - ▶ There are 3: aws (Standard), aws-cn (China), and aws-us-gov (AWS GovCloud (US))
 - ▶ A bucket is similar to Internet domain names
 - ▶ AWS currently has three partitions: Objects stored in the buckets are retrieved using a HTTP URL address from anywhere
- ▶ Example: If an object with named **"/photos/mygarden.jpg"** is stored in the **myawsbucket** bucket, then it can be accessed by:

<https://myawsbucket.s3.amazonaws.com/photos/mygarden.jpg>

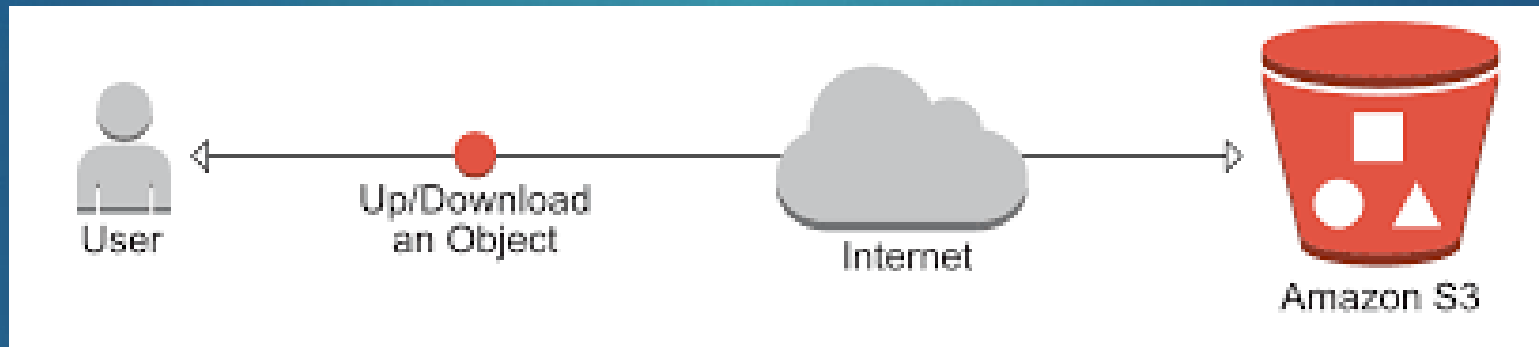
Bucket

Object

- ▶ Question – What else is important about naming a bucket?

S3 – Try it for yourself – Part 1

- ▶ Do the exercise in [Assignments > Activity – Store and Retrieve a file with S3](#)
- ▶ Don't forget to upload a screenshot when you are done



Term Project Teams

- ▶ Meet your team!
- ▶ Please join the Breakout room that pertains to your Team #
- ▶ Introduce yourself, how your team will communicate, etc.

Team 1
Hiroto Takeuchi
Kobe Laprade
Mike Andrews
Soumya Dayal
Team 2
Caleb Naeger
Leon Letournel
Tenzin Dhondup
Ty Carpenter
Team 3
Anusha Mahesh
Connor Novak
Michael Berrios
Riley Basile-Benson

Team 4
Chinmay Lokare
Jerry Lay
Poojith Panga
Christian Ashley
Team 5
Uzair Islam
Joseph Yang
Prerit Mittal
Zainab Rangwala

Storage – Elastic Block Store (EBS)

- ▶ Provides persistent block-level data storage
- ▶ A virtualized partition of a physical storage drive that's not directly connected to the EC2 instance it's associated with
- ▶ Block storage stores files in multiple volumes called “blocks”, which act as separate hard drives
 - ▶ More flexible and offer higher performance than regular file storage
- ▶ Most EC2 instances are backed with EBS storage
- ▶ Use cases include business continuity, software testing, and database management



Amazon **EBS**

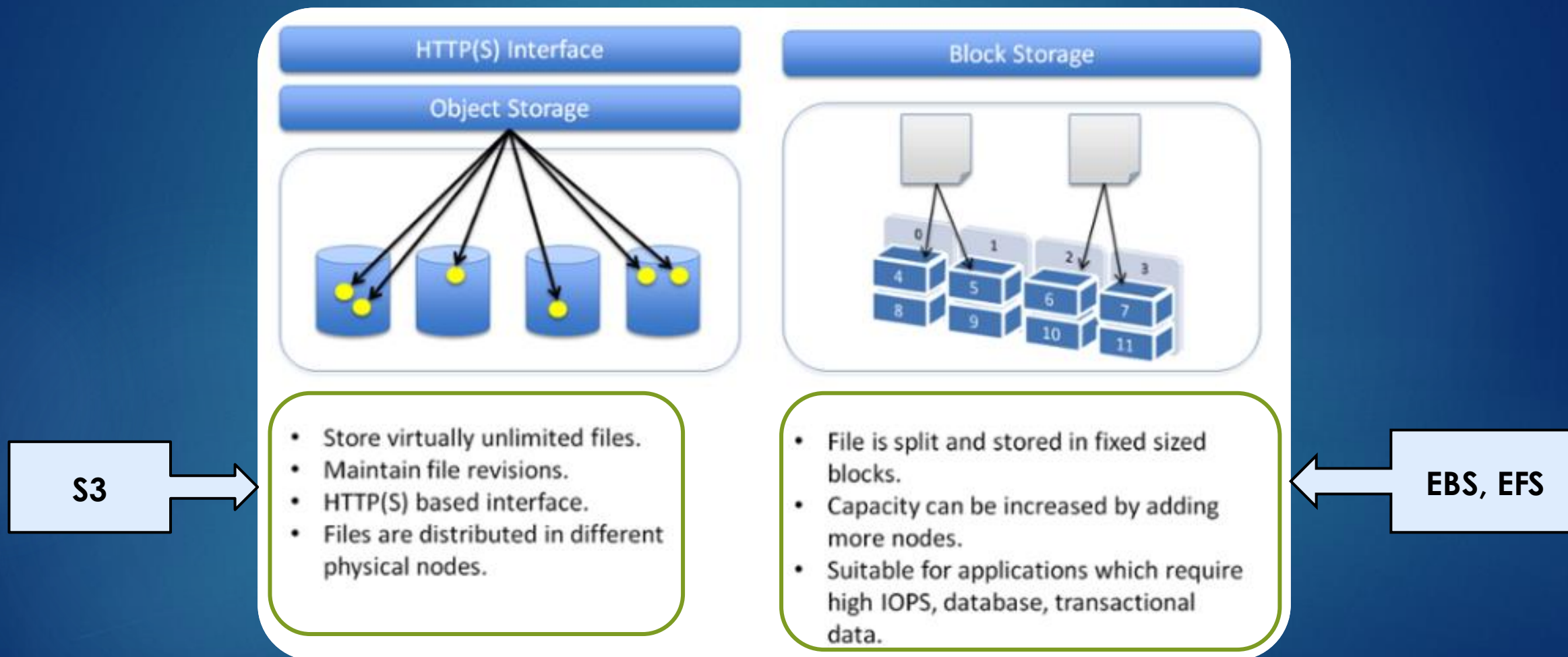
Storage – Elastic File System (EFS)

- ▶ Shared, elastic file storage system that grows and shrinks as you add and remove files
- ▶ It offers a traditional file storage paradigm, with data organized into directories and subdirectories
- ▶ EFS is useful for SaaS applications and content management systems
- ▶ You can mount EFS onto several EC2 instances at the same time



Amazon EFS

Storage – Costs/Comparisons



Storage – So what do you use?



Amazon EFS



Amazon S3



Amazon EBS

▶ Simple Storage Service (S3)

- ▶ Cheapest for data storage alone and can be accessed from anywhere via a URL
- ▶ Highly Available and redundant. 99.999999999% durability

▶ Elastic Block Store (EBS)

- ▶ Faster than S3 and EFS, more expensive than S3 but less costly than EFS
- ▶ Only available in a same region as the EC2 instance

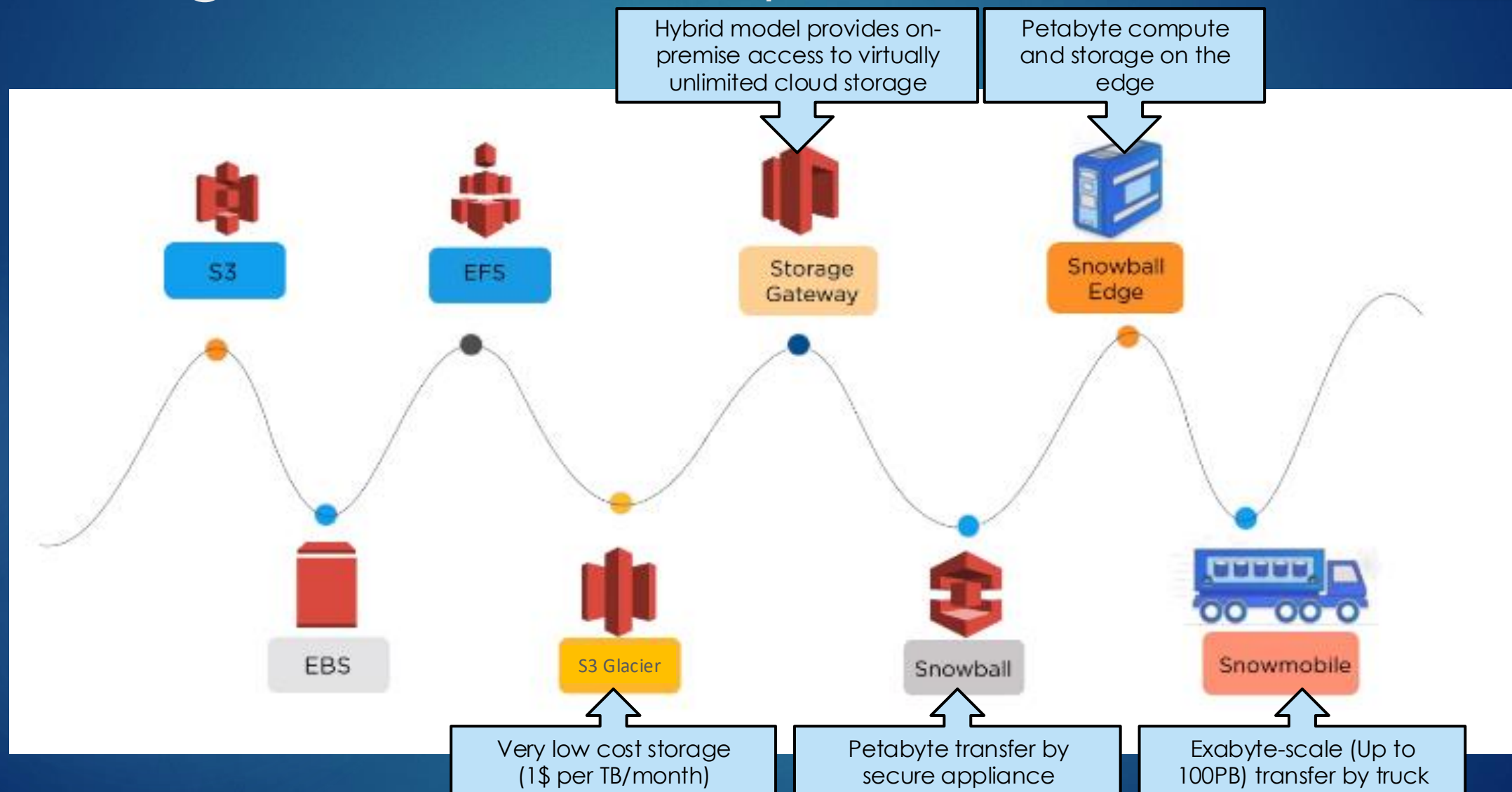
▶ Elastic File System (EFS)

- ▶ Faster than S3, most expensive compared to S3 and EBS
- ▶ Available on multiple regions and can be accessed by *thousands* on EC2 instances at once
- ▶ Best for large very large quantities of data, such as large analytic workloads, which may exceed the capabilities of EBS



Conclusion: A Balance of Cost and Necessity

Storage – Alternative options



Tagging

- ▶ Each tag is a simple label consisting of a user-defined key and an optional value
- ▶ Makes it easier to manage, search for, and filter resources for billing, environments (dev/test/prod), etc.

▼ **Name and tags** [Info](#)

Key [Info](#)

Value [Info](#)

Resource types [Info](#)

▼

49 remaining (Up to 50 tags maximum)

Specify Key Pair

- ▶ Required for accessing your EC2 instance and other AWS resources
- ▶ NOTE: If you lose your key pair or run into another other issues, you can easily create a new pair under **EC2 > Network & Security > Key Pairs**

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

▼

 [Create new key pair](#)

Advanced Options

- ▶ There are many more options under “Advanced details” to further configure your EC2 instances
- ▶ We will revisit this later

Advanced details [info](#)

Purchasing option [info](#)
☐ Request Spot Instances
Request Spot Instances at the Spot price, capped at the On-Demand price

Domain join directory [info](#)
Select [Create new directory](#)

IAM instance profile [info](#)
Select [Create new IAM profile](#)

Hostname type [info](#)
IP name

DNS Hostname [info](#)
☒ Enable IP name IPv4 (A record) DNS requests
☒ Enable resource-based IPv4 (A record) DNS requests
☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery [info](#)
Select

Shutdown behavior [info](#)
Select

Stop - Hibernate behavior [info](#)
Select

Termination protection [info](#)
Select

Stop protection [info](#)
Select

Detailed CloudWatch monitoring [info](#)
Select

Elastic GPU [info](#)
Select

Elastic Inference [info](#)
☐ Add Elastic Inference accelerators

Credit specification [info](#)
Select

Placement group name [info](#)
Select [Create new placement group](#)

EBS-optimized instance [info](#)
Disable

Capacity reservation [info](#)
Select

Tenancy [info](#)
Select

RAM disk ID [info](#)
Select

Kernel ID [info](#)
Select

Nitro Enclave [info](#)
Select
Nitro Enclaves are not compatible with instance types that have fewer than 2 vCPUs.

License configurations [info](#)
Select a license configuration [refresh](#)

☐ Specify CPU options
The selected instance type does not support CPU options.

Metadata accessible [info](#)
Select

Metadata version [info](#)
Select

Metadata response hop limit [info](#)
Select

Allow tags in metadata [info](#)
Select

User data [info](#)

☐ User data has already been base64 encoded

Accessing EC2: CloudShell vs. your PC

- ▶ CloudShell runs from your browser, and you can do use this for most activities and term project. It comes with 1 GB per region. Disk is persisted, installed software is not.

```
AWS CloudShell

us-east-1

Preparing your terminal...
[cloudshell-user@ip-10-1-79-71 ~]$ Try these commands to get started:
aws help or aws <command> help or aws <command> --cli-auto-prompt
[cloudshell-user@ip-10-1-79-71 ~]$
```

- ▶ If you prefer using your PC, there are a lot of options (Git bash, PuTTY, powershell, etc.)
 - ▶ Some easier than others to set up
- ▶ You will need to install **AWS CLI** (Command-line Interface) to access AWS
- ▶ AWS CLI is pre-installed on CloudShell

```
cloud-academy — ec2-user@ip-172-31-89-37:~ — ssh -i MZ_AWS_ACADEMY.pem ec2-user@ec2-54-236-245-192.comp...
mikez@Mikes-MacBook-Pro-3 cloud-academy % ssh -i "MZ_AWS_ACADEMY.pem" ec2-user@ec2-54-236-245-192
.compute-1.amazonaws.com
The authenticity of host 'ec2-54-236-245-192.compute-1.amazonaws.com (54.236.245.192)' can't be e
stablished.
ED25519 key fingerprint is SHA256:zMfmmOYHXjiitKb/2ilIUme761RX/jr7NIZxiDiFMQ8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-236-245-192.compute-1.amazonaws.com' (ED25519) to the list of
known hosts.

  __|  __|_  )
  _| (  _/   Amazon Linux 2 AMI
 ---|\\___|___|

https://aws.amazon.com/amazon-linux-2/
12 package(s) needed for security, out of 22 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-89-37 ~]$
```

AWS Accounts

- ▶ If you are using a new AWS service, make sure you check whether how it's covered under your account (see right)
- ▶ There will be some activities we will do this semester that will require a slight charge

Displaying 1-8 (186)

Amazon EC2

Use credits to access Amazon EC2 features in both Free and Paid plans. Free plan eligible instances include:

T3.micro, T3.small, T4g.micro, T4g.small, C7i-flex.large, M7i-flex.large

Learn more → Pricing →

Amazon S3

Use credits to access Amazon S3 features in both Free and Paid plans, including:

Storage of data across multiple zones with 99.999999999% durability

Protection of data with encryption, access control, and Block Public Access settings

Learn more → Pricing →

Amazon RDS

Use credits to access Amazon RDS features in both Free and Paid plans. Free plan eligible instances include:

db.t3.micro and db.t4g.micro and 4 engines: MySQL, PostgreSQL, MariaDB, and Microsoft SQL Server

Learn more → Pricing →

Amazon DynamoDB

Amazon DynamoDB is an always free service on the Free and Paid plan. Use your credits to evaluate beyond these monthly limits:

25 GB of storage

25 provisioned Write Capacity Units (WCU)

75 provisioned Read Capacity Units (RCU)

Learn more → Pricing →

AWS Lambda

AWS Lambda is an always free service on the Free and Paid plan. Use your credits to evaluate beyond these monthly limits:

1,000,000 free requests per month

Up to 400,000 GB-seconds or 3.2 million seconds of compute time per month

Learn more → Pricing →

AI

Available on both plans

Amazon Bedrock

The easiest way to build and scale generative AI applications with foundation models

→ +

AI-ML

Available on both plans

Amazon SageMaker AI

Machine learning for every data scientist and developer.

→ +

Databases

Available on both plans

Amazon Aurora DSQL

Fastest serverless distributed SQL database for always available applications

→ +

Show 8 more

EC2 Instances – Watch your usage!

- ▶ You will get charged even if you are using it
- ▶ Only terminate if you do not plan to use your instance any more
- ▶ If you want to save your work on your EC2 instance, you can stop it and restart later
 - ▶ You will not be charged for the instance while in a stopped state
 - ▶ You will be charged other resources that are connected to it (e.g. EBS)



Where we are now...

- ▶ You are now able to create an EC2 instance and your own S3 bucket
- ▶ Let's try something more interesting using your S3 bucket
- ▶ Your next assignment is to create a public static web site with your S3 bucket to make it accessible to anyone
- ▶ Go to myCourses **Assignments > Activity – Create a Static Web Site with S3**

Term Project Teams

- ▶ There will be 1 private channel per team on Slack that I am asking teams use for their communication
 - ▶ Look for invites very shortly
- ▶ This will help me, and the CAs monitor team progress and quickly answer any questions you may have for us

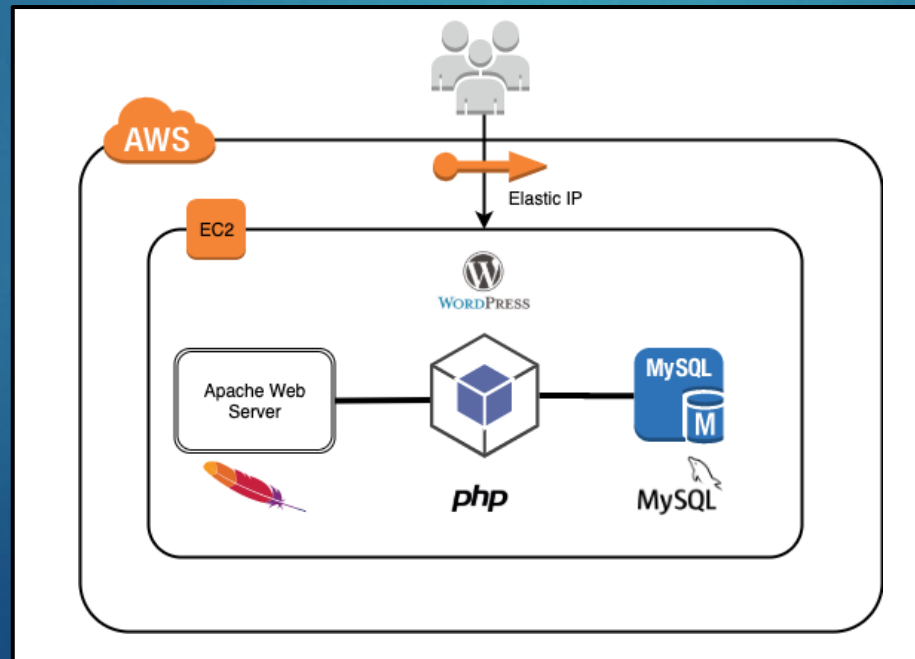


- ▶ Homework – AWS Tutorials(3 points each)
- ▶ Each team member picks and completes two AWS tutorials
 - ▶ Purpose of this is to get your team more familiar with AWS services for the term project
 - ▶ Team members can do a tutorial on the same AWS technology, but these must be different tutorials (make sure you coordinate with your team!)
- ▶ Details and guidelines can be found in myCourses under **Assignments**
> **Homework - AWS Tutorials**
 - ▶ All submissions should be posted to Discussions not Assignments
 - ▶ These will only be visible to your team, but eventually to the entire class

Due next class:

Assignments > Activity - Install WordPress on EC2

- ▶ **WordPress** is a content management system (CMS) that allows you to create and manage websites
- ▶ It's the world's most popular website builder and powers over ~43% of all websites



- ▶ Make sure you allocate enough time for this activity (20-30 minutes)
- ▶ Read the directions carefully!
- ▶ Do not do it right before next class